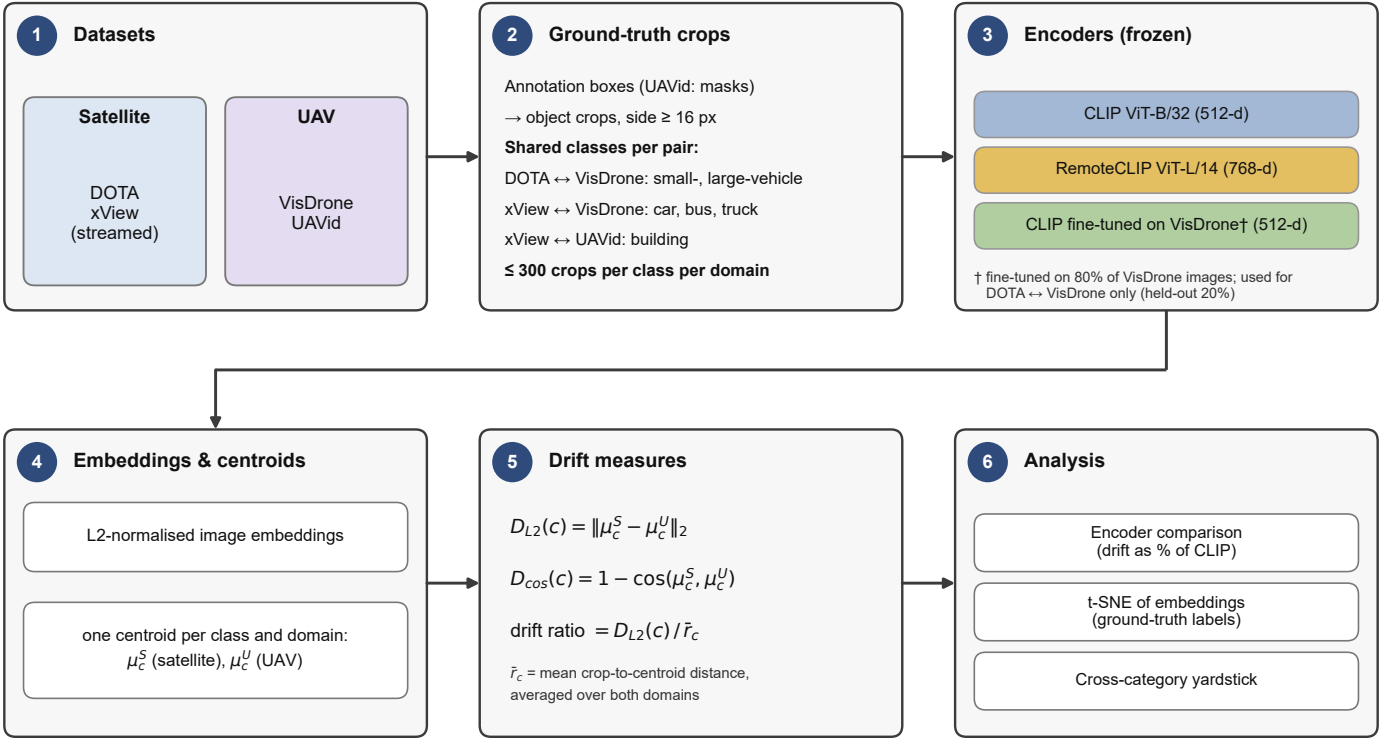

1
Datasets
Satellite
DOTA
xView
(streamed)
UAV
VisDrone
UAVid
2
Ground-truth crops
Annotation boxes (UAVid: masks)
→ object crops, side ≥ 16 px
Shared classes per pair:
DOTA ↔ VisDrone: small-, large-vehicle
xView ↔ VisDrone: car, bus, truck
xView ↔ UAVid: building
≤ 300 crops per class per domain
3
Encoders (frozen)
CLIP ViT-B/32 (512-d)
RemoteCLIP ViT-L/14 (768-d)
CLIP fine-tuned on VisDrone† (512-d)
† fine-tuned on 80% of VisDrone images; used for DOTA ↔ VisDrone only (held-out 20%)
4
Embeddings & centroids
L2-normalised image embeddings
one centroid per class and domain:
μ_c^S (satellite), μ_c^U (UAV)
5
Drift measures
$D_{L2}(c) = \|\mu_c^S - \mu_c^U\|_2$
$D_{cos}(c) = 1 - \cos(\mu_c^S, \mu_c^U)$
drift ratio $= D_{L2}(c) / \bar{r}_c$
$\bar{r}_c$ = mean crop-to-centroid distance, averaged over both domains
6
Analysis
Encoder comparison
(drift as % of CLIP)
t-SNE of embeddings
(ground-truth labels)
Cross-category yardstick